One-page Summary

We built a software tool that automatically checks whether an OTC derivatives trade meets the reporting requirements of US (CFTC) and EU (EMIR) regulations. You give it a trade, and it tells you whether it's compliant or noncompliant for both jurisdictions at once.

We made three key design choices:

1. Separate rules for each jurisdiction. US and EU require different data fields. For example, the EU demands collateral and margin information; the US does not. Our tool runs two independent checks and reports both results.

2. Product classification first. Before checking any rule, the tool asks: "What type of product is this?" It looks up the trade in the ANNADSB UPI taxonomy, a global library of derivative product definitions. If the product is not in the library, the tool stops and reports "no product definition" rather than guessing.

3. Automated validation of identifiers. LEI (Legal Entity Identifier) and UTI (Unique Transaction Identifier) have complex formatting rules. LEI includes a mathematical checksum. UTI requires the first 20 characters to match the reporting firm's own LEI.

Our tool automates these checks.Without our tool, a compliance officer works slower, cost more, and prone to mistakes. Our tool reduces a 10minute manual check to a few seconds of automated processing. It handles the routine validation so the officer can focus on complex cases.

However, it cannot handle event contracts (prediction markets, binary outcome trades). The ANNADSB taxonomy does not have an "EventContract" category. Our tool correctly reports "no product definition" for these trades, but it cannot classify or validate them. This is not a technical failure but a gap in the global classification standard that regulators and industry bodies need to address.

It cannot give legal conclusions. Compliance rules require human judgment. Our tool flags potential issues; a qualified professional must make the final determination. This reflects the principle that "judgment cannot be outsourced."

It cannot update itself when rules change. Regulatory updates require our team to modify the code. Realtime rule updates are possible in principle, but we prioritised accuracy over automation for this version.

We tested the tool on 28 simulated trades representing real derivative products, including interest rate swaps, credit default swaps, FX forwards, and event contracts.

Results come as follows, for conventional derivatives with valid data, the tool correctly returned "compliant" for both US and EU rules.

For trades with deliberate errors (wrong LEI format, mismatched UTI), the tool identified every violation.

For event contracts (T026T028), the tool reported "no product definition" and correctly applied specialcase handling: conditional reporting for US Kalshi trades, not applicable for EU.

Report

A Jurisdiction-Aware Compliance Tool for OTC Derivatives: Identifying Taxonomy Gaps Through Event Contracts

Regulatory Landscape — CFTC vs. EMIR

The US regulatory framework is anchored in the DoddFrank Act, with the CFTC implementing realtime public reporting (Part 43) and swap data recordkeeping (Part 45) requirements . In parallel, the European Market Infrastructure Regulation (EMIR) governs OTC derivatives reporting in the EU, mandating trade repositories to collect and maintain transaction data . While both regimes pursue the G20 transparency agenda, they differ in scope and granularity. EMIR requires additional collateral portfolio codes and margin data, whereas CFTC rules place greater emphasis on realtime public dissemination. Staszewska (2024) provides a comprehensive comparison, noting that these differences create operational complexity for crossjurisdictional firms . A key element is the Unique Transaction Identifier (UTI), which must follow the CPMIIOSCO waterfall methodology across both regimes .

System Architecture

Our tool implements a threelayer architecture. Member A's parser normalises raw trade data into a structured JSON schema. Member B's UPI lookup engine queries the ANNADSB taxonomy, classifying each trade as FOUND (product has a UPI template), NOT_FOUND (template missing but product category exists), or NO_PRODUCT_DEFINITION (product entirely outside taxonomy). Member C's compliance checker then applies CFTC and EMIR rules separately, producing jurisdictionaware outputs. This modular design allows independent updates to each layer — critical for adapting to regulatory changes.

Portfolio Findings

Of 28 test trades, only three achieved full CFTC compliance (T013, T017, T021), while a single trade (T013) satisfied EMIR requirements. The most frequent violation — INVALID_LEI_OTHER (24 occurrences) — reflects widespread LEI formatting and checksum errors in our test dataset. Critically, UPI_NOT_FOUND affected 24 trades, including 12 conventional OTC products lacking ANNADSB templates and three event contracts (T026T028) with NO_PRODUCT_DEFINITION status. The divergence in T026T028 outcomes is striking: CFTC treats them as CONDITIONAL (traded on Kalshi, a registered DCM), while EMIR deems them NOT_APPLICABLE (classified as gambling under German GlüStV 2021).

Policy analysis: Why Event Contracts Are Not in the Taxonomy

Economic Function Test

A threshold question for any instrument claiming derivative status is whether it satisfies an economic function test. The UPI taxonomy, maintained by ANNADSB under the oversight of the CPMIIOSCO steering group, includes product types that serve identifiable economic purposes — hedging, price discovery, and efficient risk transfer. Event contracts challenge this framework.

Identifiable economic actor. Speculators, hedgers, and market makers participate in event contract markets. Corporate treasuries face real exposure to political outcomes — tariff decisions, election results, and regulatory changes. A multinational importing from China, for example, cannot hedge a sudden tariff increase using conventional FX forwards or commodity swaps. Event contracts on trade policy outcomes provide an economic hedge where traditional derivatives do not reach.

Hedge of real exposure. The fact that political risk is difficult to hedge does not make it unhedgeable in principle. Yang (2026) analyses 291,309 event contracts across six platforms, including the CFTCregulated DCM Kalshi, and demonstrates that contract prices embed a systematic pricing wedge ($\lambda = 0.183$, $p < 10^{-15}$). This wedge indicates that event contracts behave as riskadjusted derivative instruments, not as pure probability aggregation mechanisms. Beylin (2025) argues to the contrary, warning that CFTC authorisation of binary options has drifted from the statutory goals of hedging and price discovery toward products that serve no identifiable economic purpose beyond speculation. The debate itself confirms that event contracts occupy a contested but economically meaningful space.

Price discovery. Prediction markets have demonstrated consistent accuracy. The Iowa Electronic Markets outperform professional polls. Kalshi's election contracts trade at prices that correlate with polling averages while responding faster to new information. Yang (2026) finds that the systematic pricing wedge is stable across event categories, suggesting that event contract markets develop consistent risk pricing mechanisms — a hallmark of mature derivative markets.

Conclusion. Event contracts satisfy the economic function test for derivativelike instruments. Their absence from the ANNADSB UPI taxonomy is not a reflection of their economic nature but a historical and political contingency. The taxonomy was designed around conventional asset classes — Rates, Credit, Equity, FX, Commodities — before event contracts achieved regulatory recognition and market scale.

Proposed UPI Schema Extension

If event contracts are to be brought within the UPI framework, a new schema is required. The following proposal prioritises reportability over perfection — the ability for regulators across jurisdictions to see the same trade, the same event, and the same counterparty.

Jurisdictional Arbitrage: US vs. EU

The absence of a UPI for event contracts does not mean these instruments do not trade. It means they trade in a regulatory gap. Our test suite includes three event contracts with materially different outcomes across jurisdictions.

Kalshi is a registered Designated Contract Market (DCM) under CFTC oversight. Event contracts traded on Kalshi are lawful in the US, with conditional reporting obligations under Part 43/45. In the EU, the same instrument is classified as gambling under the German Interstate Treaty on Gambling (GlüStV 2021) framework and is not subject to EMIR trade reporting. Polymarket, an offshore decentralised platform accessible via VPN, operates in neither jurisdiction's formal regulatory perimeter.

This divergence creates a classic regulatory arbitrage opportunity. A sophisticated user can access Kalshi directly (US legal), Polymarket via VPN (unregulated), or both. The compliance engine sees T027 as NOT_APPLICABLE in both regimes — a technically correct output that masks the underlying risk. The user's position across platforms is not aggregated. Consumer protection collapses when disputes arise on offshore platforms. Systemic visibility disappears because no reporting obligation attaches to either jurisdiction's "not applicable" determination.

Regulatory arbitrage is not a theoretical risk. As Holland & Knight (2025) document, the CFTC voluntarily dismissed its enforcement action against Kalshi following a district court ruling that CEA jurisdiction preempts state gambling laws. The patchwork of federal, state, and international treatment ensures that event contracts will continue to migrate toward the most permissive venue.

Limits of the Compliance Engine

Our tool implements jurisdictionaware logic. It distinguishes CFTC from EMIR requirements. It validates LEI checksums and UTI namespaces. But when a trade arrives with asset_class = "EventContract", the engine does not mismeasure risk — it fails to see the trade at all.

Member B's UPI lookup returns NO_PRODUCT_DEFINITION. Member C's compliance checker reads that status and applies specialcase handling: CONDITIONAL for Kalshi trades under CFTC, NOT_APPLICABLE for all event contracts under EMIR. The engine performs exactly as designed. The problem is not in the engine. The problem is in the taxonomy the engine queries.

The absence of UPI is not a neutral technical gap. It is a systemic risk indicator. When a growing volume of trades falls outside the classification framework, that volume does not disappear — it becomes invisible. Invisible positions cannot be margined, netted, or resolved in default. The compliance engine that cannot see an event contract is not a failed engine. It is a mirror held up to a classification regime that has not yet caught up with the instruments it is meant to govern.

Limitations and Future Work

Our tool has three principal limitations. First, it cannot classify event contracts because the ANNADSB taxonomy lacks an EventContract asset class. Second, rule updates require manual code changes — a risk when regulations change midperiod. Third, the tool only supports CFTC and EMIR; other jurisdictions remain unimplemented. With additional resources, we would prioritise APIbased rule versioning and a formal UPI extension proposal to ANNADSB.

Team Contributions

Member A Zhang Meng: Implemented the Module 1 parser and data pipeline for OTC trade ingestion. Responsible for JSON parsing, schema validation, missing field detection, timestamp/date validation, trade normalization, and classification of conventional versus novel instruments. Generated structured outputs for downstream UPI lookup and compliance modules, including handling of prediction/event contracts outside the current ANNADSB taxonomy.

Member B Liu Tianyu: The UPI query engine for Module 2 was implemented, including dynamic product template matching and attribute validation based on the ANNADSB code set, and the content of this module in the paper.

Member C Sun Yifan： Implemented a multi-jurisdictional compliance engine that validates 28 trades against CFTC (mandatory) and EMIR (optional) reporting rules. My module performs ISO 7064 MOD 97-10 LEI checks, UTI namespace validation, and verifies all mandatory fields (UTI, LEIs, timestamps, notional, UPI, cleared indicator) while correctly handling special statuses (CONDITIONAL / NOT_APPLICABLE) for prediction contracts T026-T028. The output is a structured JSON report detailing each trade's compliance status and violations, which served as the foundation for the team's policy analysis and final presentation.

Member D Liu Guobang: Authored this report, including regulatory landscape, portfolio findings, policy analysis, and limits of compliance engine.

References:

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